KAIZEN MATHS

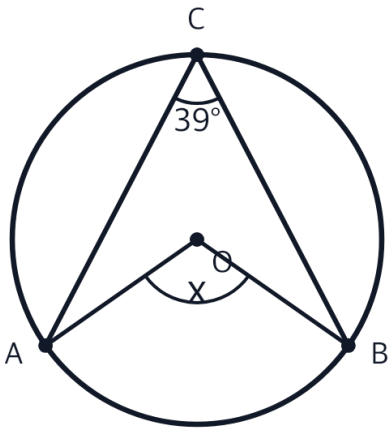
Lesson starter

Circle Theorems

Core Theorems

Name _____

SA1



In the circle, ∠ACB = 39°. Find ∠AOB.

GEO-475F-L1 · SA

Show your working

KAIZEN MATHS

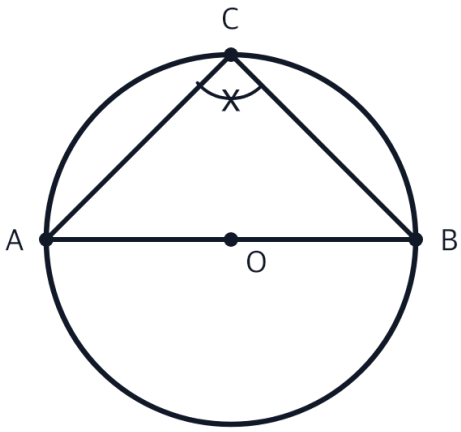
Lesson starter

Circle Theorems

Core Theorems

Name _____

SB1



AB is a diameter. Find ∠ACB.

KAIZEN MATHS

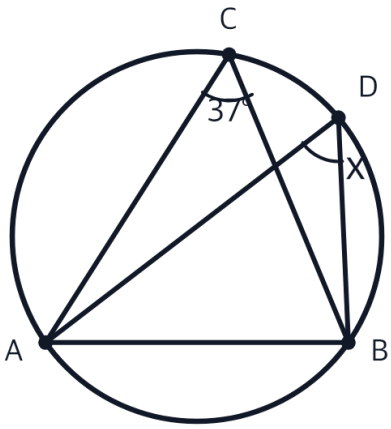
Lesson starter

Circle Theorems

Core Theorems

Name _____

SC1



Angles ∠ACB and ∠ADB stand on chord AB. If ∠ACB = 37°, find ∠ADB.

GEO-475F-L1 · SC

Show your working

KAIZEN MATHS

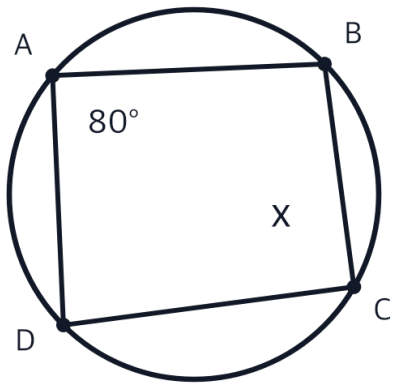
Lesson starter

Circle Theorems

Core Theorems

Name _____

SD1



ABCD is a cyclic quadrilateral. If ∠DAB = 80°, find ∠BCD.

GEO-475F-L1 · SD

Show your working

KAIZEN MATHS

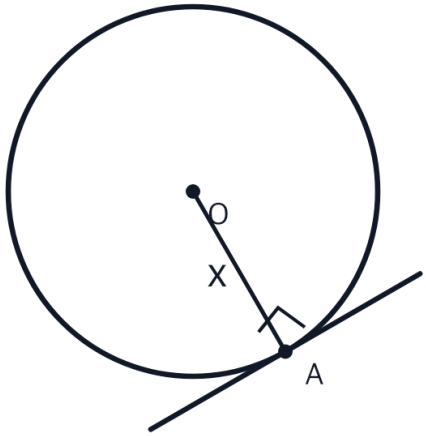
Exit ticket

Circle Theorems

Core Theorems

Name _____

EA1



The line through A is a tangent. Find the angle between the tangent and radius OA.

GEO-475F-L1 · EA

Show your working

KAIZEN MATHS

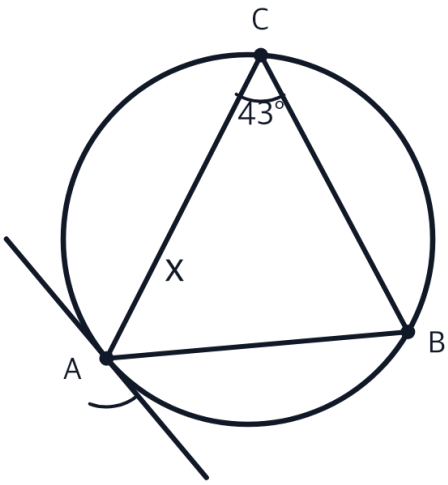
Exit ticket

Circle Theorems

Core Theorems

Name _____

EB1



The line at A is a tangent and ∠ACB = 43°. Find the angle between the tangent and chord AB.

GEO-475F-L1 · EB

Show your working

KAIZEN MATHS

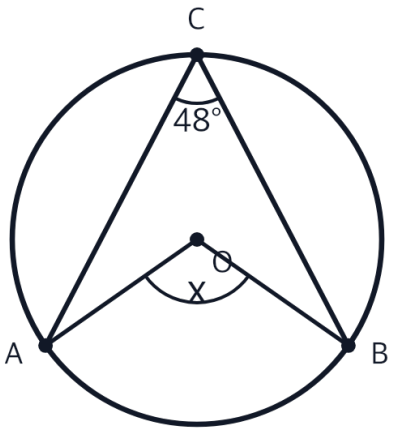
Exit ticket

Circle Theorems

Core Theorems

Name _____

EC1



In the circle, ∠ACB = 48°. Find ∠AOB.

GEO-475F-L1 · EC

Show your working

KAIZEN MATHS

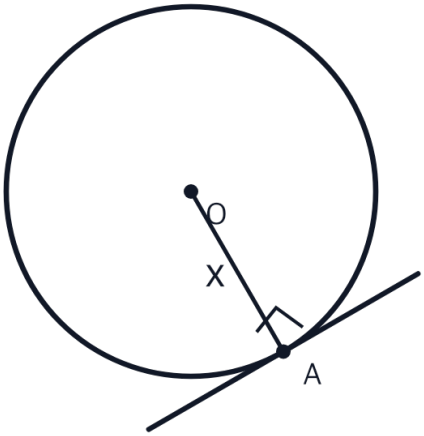
Exit ticket

Circle Theorems

Core Theorems

Name _____

EA1



The line through A is a tangent. Find the angle between the tangent and radius OA.

GEO-475F-L1 · EA

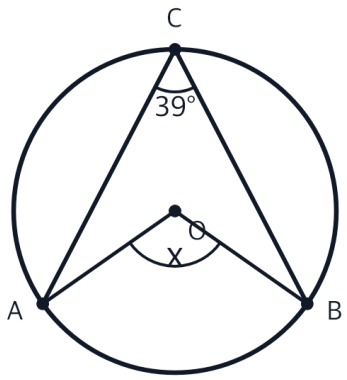
Show your working

Teacher answers & working

Circle Theorems · Core Theorems

GEO-475F-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



In the circle, ∠ACB = 39°. Find ∠AOB.

Answer:

78°

Working:

∠ACB and ∠AOB stand on the same arc AB.

The angle at the centre is twice the angle at the circumference.

$\angle AOB = 2 \times 39°$

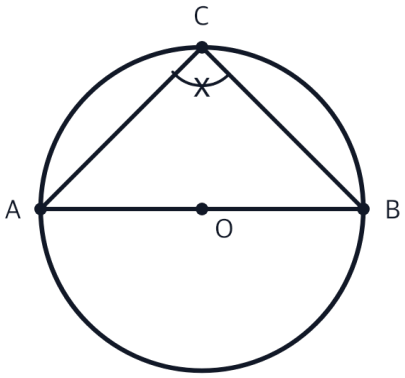
$\angle AOB = 78°$

Teacher answers & working

Circle Theorems · Core Theorems

GEO-475F-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



AB is a diameter. Find ∠ACB.

Answer:

90°

Working:

AB is a diameter, so arc AB is a semicircle.

The angle in a semicircle is a right angle.

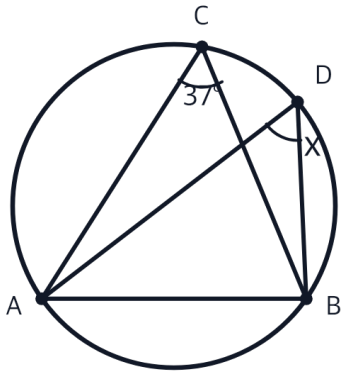
$\angle ACB = 90°$

Teacher answers & working

Circle Theorems · Core Theorems

GEO-475F-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



Angles ∠ACB and ∠ADB stand on chord AB. If ∠ACB = 37°, find ∠ADB.

Answer:

37°

Working:

Both ∠ACB and ∠ADB stand on chord AB.

Angles in the same segment are equal.

$\angle ADB = \angle ACB$

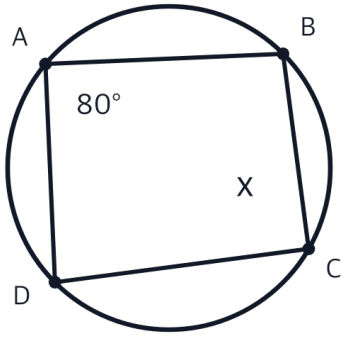
$\angle ADB = 37°$

Teacher answers & working

Circle Theorems · Core Theorems

GEO-475F-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



ABCD is a cyclic quadrilateral. If ∠DAB = 80°, find ∠BCD.

Answer:

100°

Working:

Opposite angles in a cyclic quadrilateral add to 180°.

$\angle DAB + \angle BCD = 180°$

$80° + \angle BCD = 180°$

$\angle BCD = 180° - 80°$

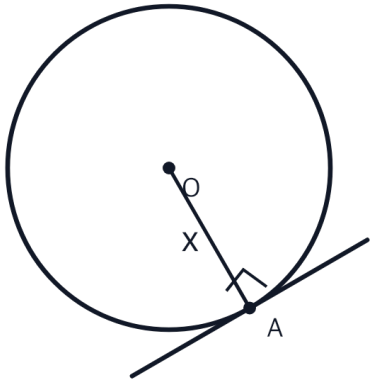
$\angle BCD = 100°$

Teacher answers & working

Circle Theorems · Core Theorems

GEO-475F-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



The line through A is a tangent. Find the angle between the tangent and radius OA.

Answer:

90°

Working:

A tangent is perpendicular to the radius at the point of contact.

The tangent touches the circle at A and OA is a radius.

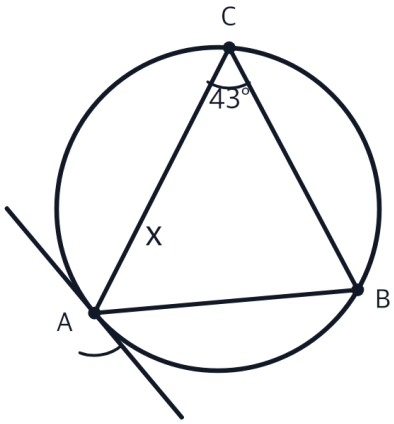
Therefore the angle between the tangent and OA is 90°.

Teacher answers & working

Circle Theorems · Core Theorems

GEO-475F-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



The line at A is a tangent and ∠ACB = 43°. Find the angle between the tangent and chord AB.

Answer:

43°

Working:

The angle between a tangent and a chord equals the angle in the alternate segment.

The angle in the alternate segment is ∠ACB.

So the tangent-chord angle equals ∠ACB.

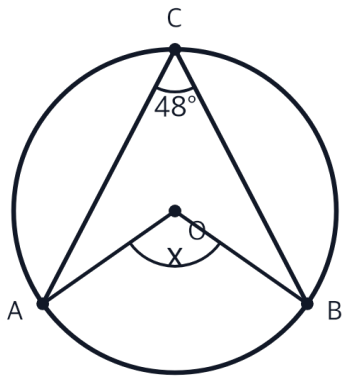
$x = 43°$

Teacher answers & working

Circle Theorems · Core Theorems

GEO-475F-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1



In the circle, ∠ACB = 48°. Find ∠AOB.

Answer:

96°

Working:

∠ACB and ∠AOB stand on the same arc AB.

The angle at the centre is twice the angle at the circumference.

$\angle AOB = 2 \times 48°$

$\angle AOB = 96°$